

1.1. WHAT IS ORACLE?

Oracle database is a relational database management system (RDBMS) from Oracle Corporation. The RDBMS system manages the relational data. Oracle Database is the most famous relational database system (RDBMS) because it shares the largest part of a market among other relational databases. Some other popular relational databases are MySQL, DB2, SQL Server, PostgreSQL, etc.

Oracle database is a relational database management system. It is also called **OracleDB**, or simply **Oracle**. It is produced and marketed by **Oracle Corporation**. It was created in 1977 by **Lawrence Ellison** and other engineers. It is one of the most popular relational database engines in the IT market for storing, organizing, and retrieving data.

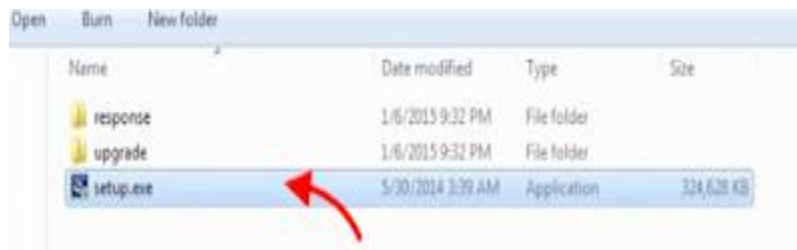
1.2. Editions of Oracle database

Oracle database is compatible with a wide range of platforms such as Windows, UNIX, Linux, and macOS. The following is a list of Oracle database editions in order of priority:

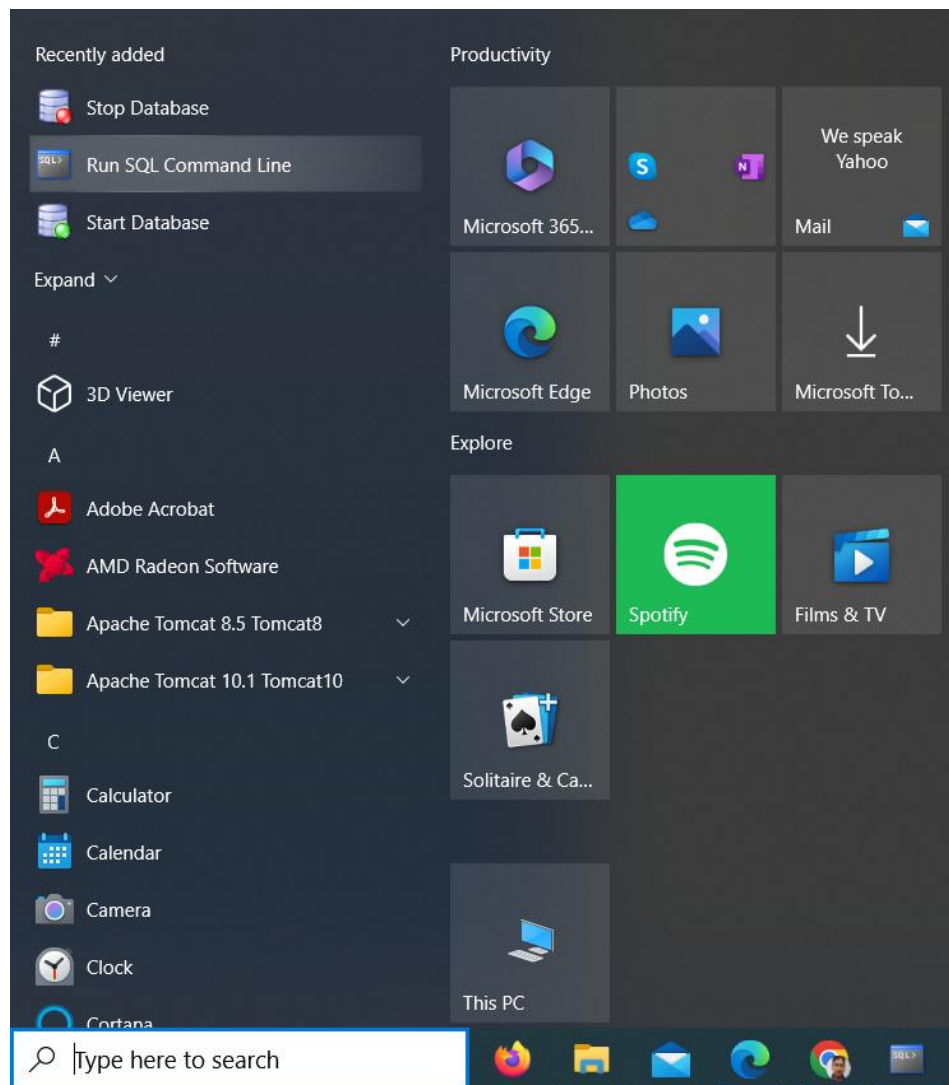
- **Enterprise Edition:** It is the most robust and secure edition. It offers all features, including superior performance and security.
- **Standard Edition:** It provides the base functionality for users that do not require Enterprise Edition's robust package.
- **Express Edition (XE):** It is the lightweight, free and limited Windows, and Linux edition.
- **Oracle Lite:** It is designed for mobile devices.
- **Personal Edition:** It's comparable to the Enterprise Edition but without the Oracle Real Application Clusters feature.

1.3. Install and Open11g XE(Express Edition)

- Run setup.exe



- Click Next
- Accept the license agreement and click Next
- Configure the password for the database. Save this password because you need it to connect with database.(This password will be used later to connect system database in our case it will be “admin”)
- Save the summary. This is the connection details to connect with the database. We will use this to configuration details in SQL developer tool to connect with this database in below steps.
- Database installation is successful. Now the database is ready to use.
- Mostly database get started automatically after installation. But as this is the first time let's start the database manually.
- Search for the Oracle Database 11g XE and click on Run SQL Command Line.



- In SQL Command Line type Connect System and enter Password “admin”

```
SQL> connect system
Enter password:
Connected.
SQL> connect system
Enter password:
Connected.
```

SQL, which is short for Structured Query Language, although some places call it Server Query Language, is the database language which you use to access data stored in your Oracle Database. SQL as a data retrieval language, is an industry standard and all relational database products, not only Oracle, implement SQL as the mechanism, the language the syntax you use to retrieve data from your database server.

1.4. Oracle DATABASE LANGUAGES

A database system provides a data definition language to specify the database schema and a data manipulation language to express database queries and updates and data control language to control the specific task. These languages are:

1. Data definition language (DDL)
2. Data manipulation language (DML).
3. Data control language (DCL).
4. Transaction control (TCL).

1.4.1. Data definition language (DDL)

Data Definition Language (DDL) statements are used to define the database structure or schema.

Some examples:

- **CREATE** - Used to create objects in the database.
- **ALTER** - Alters the structure of the database.

- **TRUNCATE** - Remove all records from a table, including all spaces allocated for the records are removed.
- **DROP** - Delete objects from the database.

1. CREATE. This command is used to create objects such as table or view in the database.

Syntax:

CREATE TABLE table_name (column1 datatype [NULL | NOT NULL], column2 datatype [NULL | NOT NULL], ... column_n datatype [NULL | NOT NULL]);

- table_name: It specifies the name of the table which you want to create.
- column1, column2, ... column n: It specifies the columns which you want to add in the table. Every column must have a data type. Every column should either be defined as "NULL" or "NOT NULL". In the case, the value is left blank; it is treated as "NULL" as default.

To create a table named student, which contain the information about the student we use the create command as shown in Fig. 1.1. For Example to create student table:

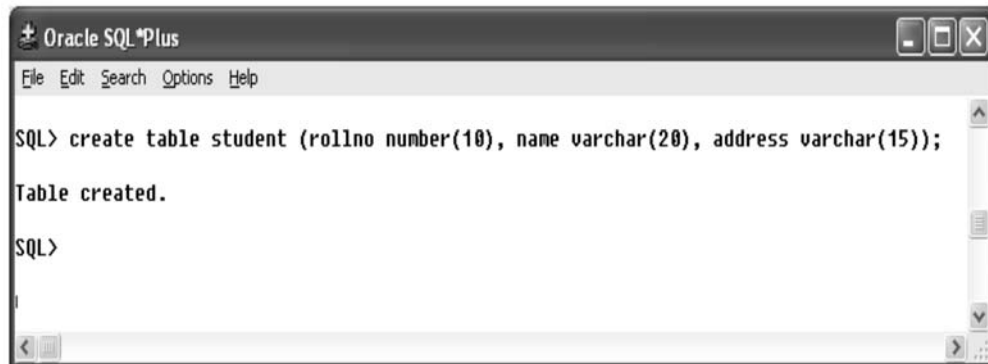


Fig. 1.1. Create Command.

CREATE TABLE customers (customer_id number(10) NOT NULL, customer_name varchar2(50) NOT NULL, city varchar2(50));

- **customer_id:** It is the first column created as a number datatype (maximum 10 digits in length) and cannot contain null values.
- **customer_name:** it is the second column created as a varchar2 datatype (50 maximum characters in length) and cannot contain null values.
- **city:** This is the third column created as a varchar2 datatype. It can contain null values.

We can CREATE TABLE with primary key. A primary key is a single field or combination of fields that contains a unique record. It must be filled. None of the field of primary key can contain a null value. A table can have only one primary key.

CREATE TABLE customers (customer_id number(10) NOT NULL, customer_name varchar2(50) NOT NULL, city varchar2(50), **CONSTRAINT** customers_pk **PRIMARY KEY** (customer_id));

```
SQL> CREATE TABLE custom1 ( customer_id number(10) NOT NULL, customer_name varchar2(50) NOT NULL, city varchar2(50),
  CONSTRAINT customers_pk PRIMARY KEY (customer_id) );
Table created.
SQL> desc custom1;
Name                               Null?    Type
-----
CUSTOMER_ID                        NOT NULL NUMBER(10)
CUSTOMER_NAME                      NOT NULL VARCHAR2(50)
CITY                              VARCHAR2(50)
```

2. ALTER - This command, alters the structure of the database. We can alter the structure of the table by using add and modify command.

(i) **Add** -: This command, is used to add new field in the table. In student table I am adding a new field named class as shown in the Fig. 9.2.

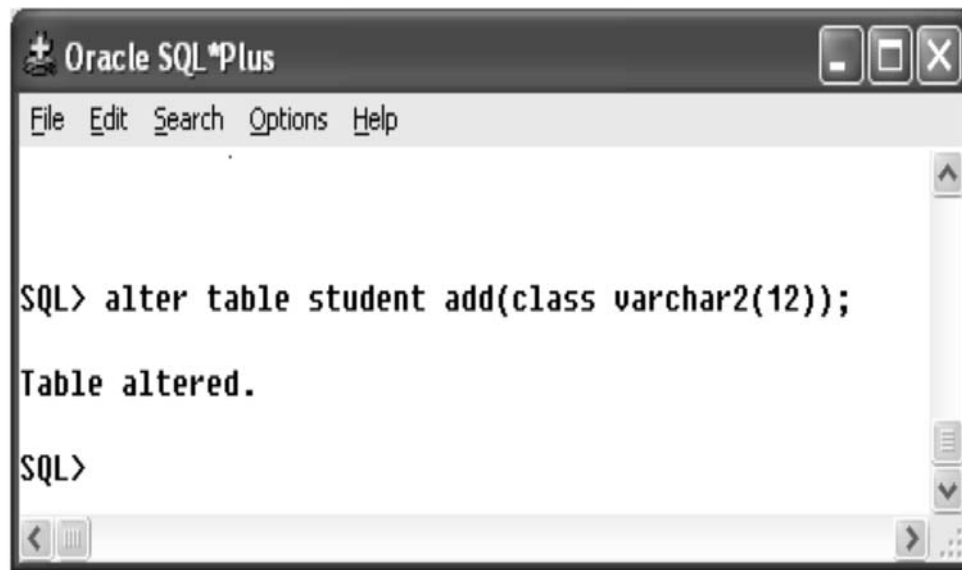


Fig. 9.2. Adding New Field in Table.

(ii) **Modify** -: This command, is used to modify the size of existing field of the table. Here in student table I am changing the size of rollno field from 10 to 8 as shown in Fig. 9.3.

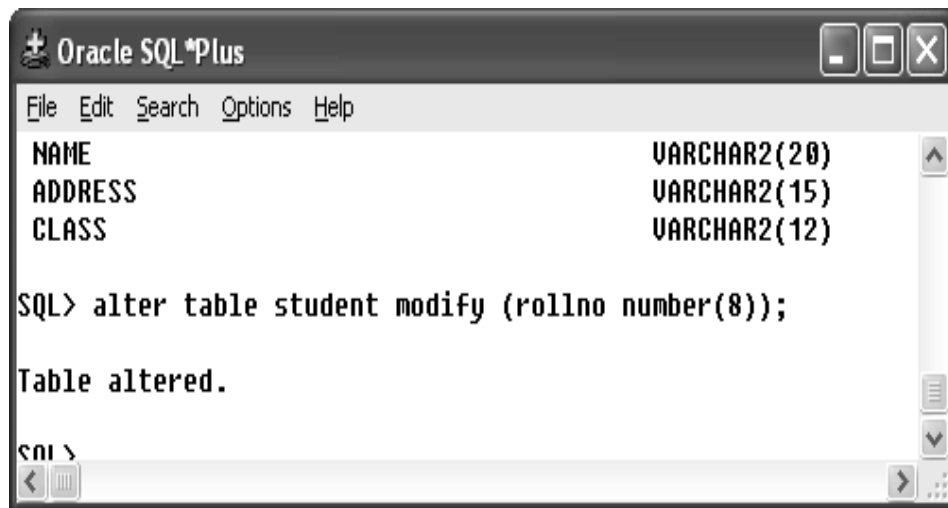


Fig. 9.3. Alter the Table with Modify Command.

(a) **TRUNCATE.** This command, removes all records from a table, including all spaces allocated for the records are removed. To remove all the records of student database use the "truncate student" command as shown in Fig. 9.4.

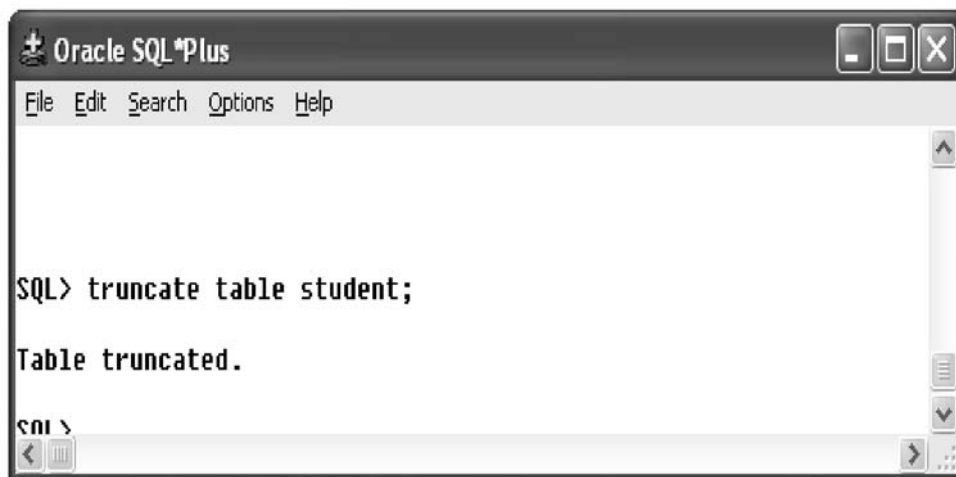


Fig. 9.4. Truncate the Table.

(b) **DROP.** This command, is used to delete objects from the database. To drop the student database use the following command as shown in Fig. 9.5.



Fig. 9.5. Drop the Table.

1. Data manipulation language (DML). Data Manipulation Language (DML) statements are used for managing data within schema objects. Some examples:

- (a) **INSERT** - Insert data into a table.
- (b) **SELECT** - Retrieve data from the a database.
- (c) **UPDATE** - Updates existing data within a table.
- (d) **DELETE** - Deletes all records from a table, the space for the records remain.

(a) **INSERT**. This command, is used to insert data into a table. Let insert a new record in student table as shown in Fig. 9.6.

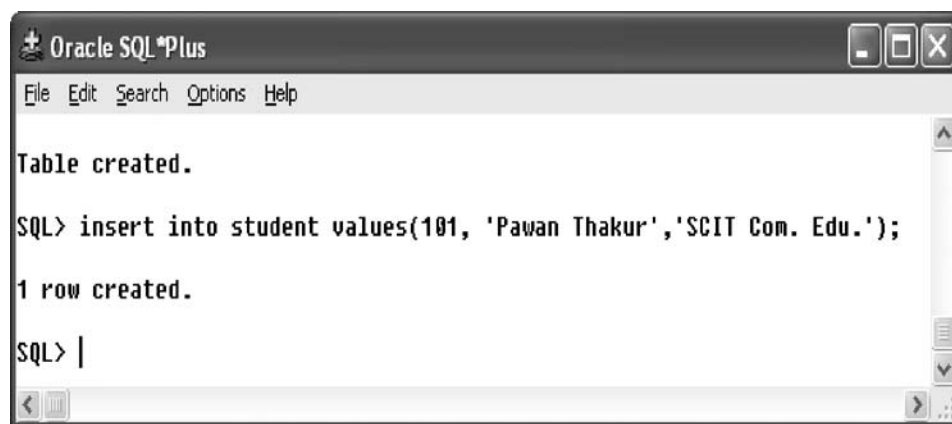
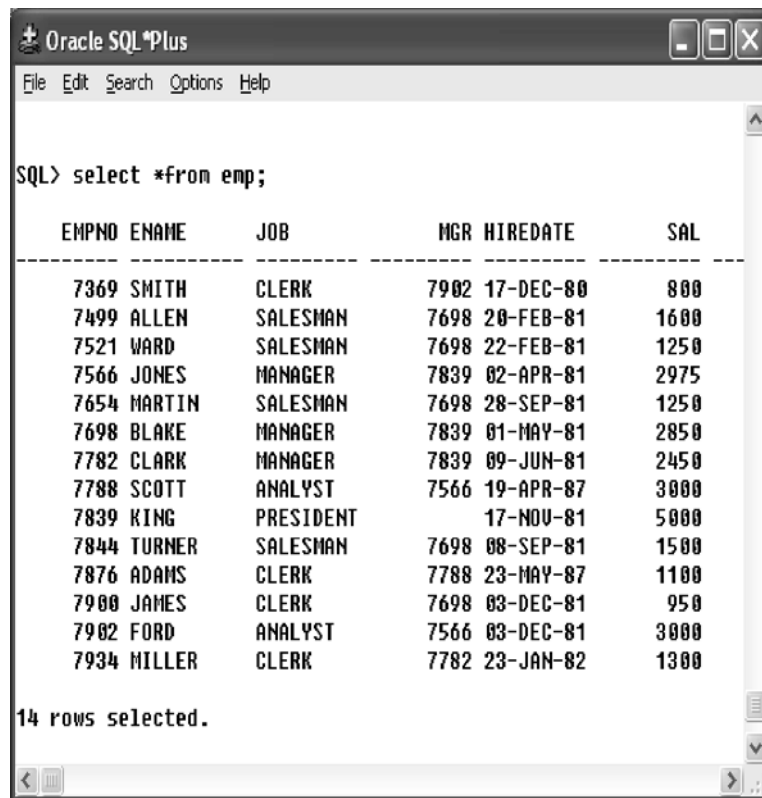


Fig. 9.6. Insert Command.

(b) **SELECT**. This command, is used to retrieve data from the database For Example to display all the records of emp table use the following command as shown in the Fig.

9.7.



The screenshot shows the Oracle SQL*Plus interface. The command prompt shows 'SQL> select *from emp;'. Below the command, a table of employee data is displayed. The table has columns: EMPNO, ENAME, JOB, MGR, HIREDATE, and SAL. There are 14 rows of data. At the bottom, it says '14 rows selected.'

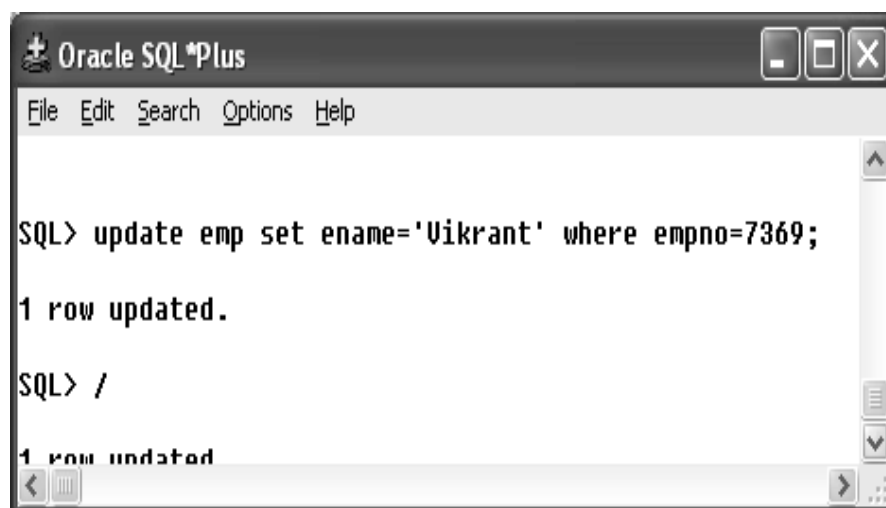
```
SQL> select *from emp;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL
7369	SMITH	CLERK	7902	17-DEC-80	800
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600
7521	WARD	SALESMAN	7698	22-FEB-81	1250
7566	JONES	MANAGER	7839	02-APR-81	2975
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250
7698	BLAKE	MANAGER	7839	01-MAY-81	2850
7782	CLARK	MANAGER	7839	09-JUN-81	2450
7788	SCOTT	ANALYST	7566	19-APR-87	3000
7839	KING	PRESIDENT		17-NOV-81	5000
7844	TURNER	SALESMAN	7698	08-SEP-81	1500
7876	ADAMS	CLERK	7788	23-MAY-87	1100
7900	JAMES	CLERK	7698	03-DEC-81	950
7902	FORD	ANALYST	7566	03-DEC-81	3000
7934	MILLER	CLERK	7782	23-JAN-82	1300

14 rows selected.

Fig. 9.7. Select Command.

(c) **UPDATE.** This command, is used to update existing data within a table. For example : the following statement update the name of emp with “Vikrant” where the empno=7369as shown in Fig. 9.8.



The screenshot shows the Oracle SQL*Plus interface. The command prompt shows 'SQL> update emp set ename='Vikrant' where empno=7369;'. Below the command, it says '1 row updated.'. Then the prompt shows 'SQL> /' and at the bottom, it says '1 row updated'.

```
SQL> update emp set ename='Vikrant' where empno=7369;
```

1 row updated.

```
SQL> /
```

1 row updated

Fig. 9.8. Update command.

update emp set ename='Vikrant' where empno=7369;

(d) **DELETE.** This command, is used to delete records from a table, the space for therecords remain. For example: To the records from emp table whose name is MILLER we will use the following statement as shown in Fig. 9.9.

Delete emp where ename ='MILLER';



Fig. 9.9. Delete a Record.

2. Data control language (DCL). The data control language (DCL) statements are used to control the data. They include:

- (a) **GRANT.** This command, is used to give user's access privileges to database.
- (b) **REVOKE.** This command, is used to withdraw access privileges given with the GRANT command.

3. Transaction control (TCL). Transaction control (TCL) statements, are used to manage the changes made by DML statements. It allows statements to be grouped together into logical transactions.

- (a) **COMMIT.** This command is used to save work done.
- (b) **SAVEPOINT.** This command is used to identify a point in a transaction to which you can later roll back.
- (c) **ROLLBACK.** This command is used to restore database to original since the last COMMIT.